The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 January 2007.

INCH-POUND

MIL-PRF-19500/286G
10 October 2006
SUPERSEDING
MIL-PRF-19500/286F
3 October 2002

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, TYPES 1N4245 THROUGH 1N4249, 1N4245EG1 THROUGH 1N4249EG1, JAN, JANTX, JANTXV, AND JANHC

These devices are inactive for new design, preferred devices are - 1N5614, 1N5616, 1N5618, 1N5620, 1N5622 on MIL-PRF-19500/427.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, power rectifier, medium-recovery diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, figure 2, and figure 3 epoxy-glass case outline.

* 1.3 Maximum ratings.

Type	V_R	V_{RWM}	I_O (1) T_A = 55°C (2)	I_{FSM} I_O = 1.0 A dc t_p = 8 ms
	V dc	V (pk)	A dc	A (pk)
1N4245, EG1 (3)	200	200	1.0	25
1N4246, EG1 (3)	400	400	1.0	25
1N4247, EG1 (3)	600	600	1.0	25
1N4248, EG1 (3)	800	800	1.0	25
1N4249, EG1 (3)	1,000	1,000	1.0	25

Types	Barometric pressure (reduced)	T_{STG} and T_J	$R_{\Theta JL}$	$Z_{\theta JX}$	t_{rr}
	mm Hg	°C	°C/W	°C/W	µs
1N4245, EG1 (3)	8	-65 to +175	42	4.5	5.0
1N4246, EG1 (3)	8	-65 to +175	42	4.5	5.0
1N4247, EG1 (3)	8	-65 to +175	42	4.5	5.0
1N4248, EG1 (3)	33	-65 to +175	42	4.5	5.0
1N4249, EG1 (3)	33	-65 to +175	42	4.5	5.0

See notes on next page.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 5961.

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1.3 Maximum ratings - continued.

(1) I_O rating is independent of heat sinking, special mounting, or forced air across the body or leads of the device.
(2) Derate linearly at 8.33 mA/°C between T_A = +55°C to +175°C.
(3) See note in 3.8 herein.

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil. or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

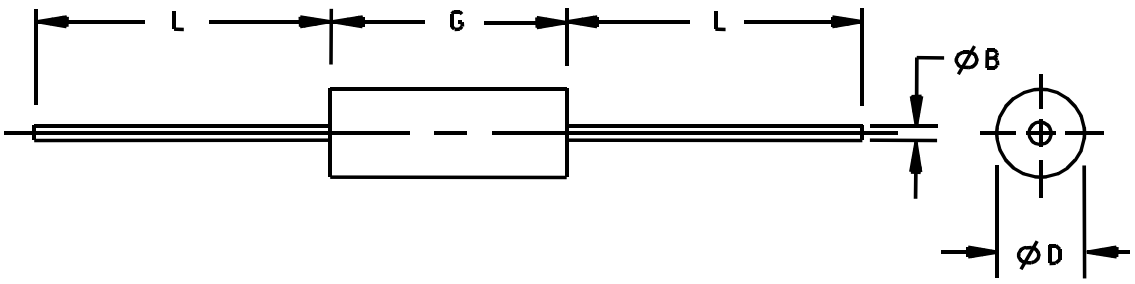
3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

EG - Epoxy over glass hermetically sealed EG1 case.

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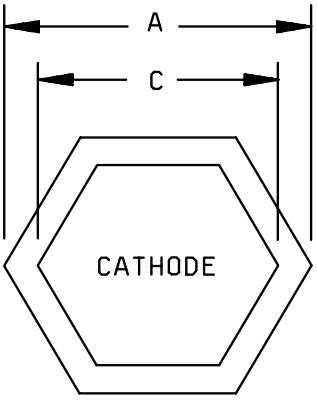


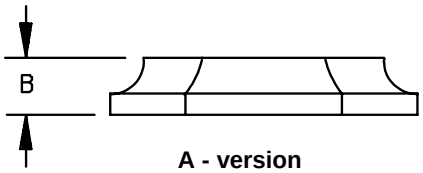
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.027	.033	0.69	0.84	
ϕD	.065	.150	1.65	3.81	3
G	.140	.250	3.56	6.35	
L	1.000	1.500	25.40	38.10	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension ϕD shall be measured at the largest diameter.
4. The G dimension shall include all uncontrolled areas of the device leads.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕX symbology.

FIGURE 1. Physical dimensions.

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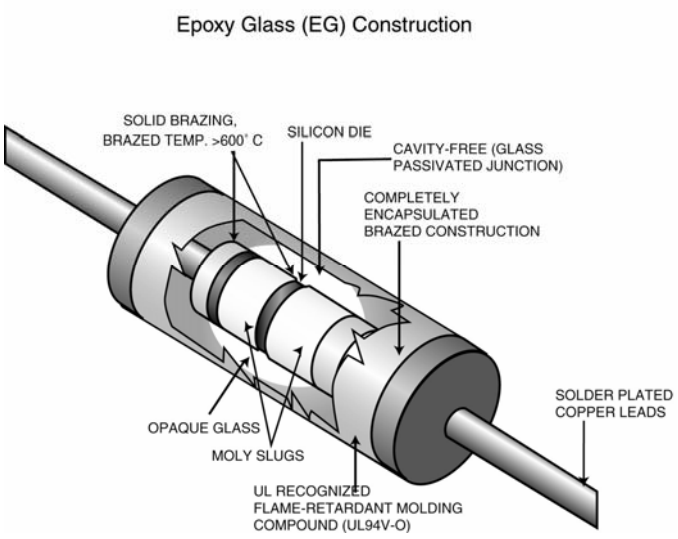
A - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕX symbology.

FIGURE 2. JANHC (A-version) die dimensions.

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NOTE:
1. See note in 3.8.
2. See figure 1 for dimensions.

FIGURE 3. Epoxy-Glass (EG1 case outline) construction (for reference only).

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3.4 Diode construction. Except for the EG1 version and the JANHC devices, these devices shall be constructed utilizing metallurgically bonded, thermally matched, non-cavity, double-plug construction. The metallurgical bond between the chip and the plug shall meet the category I requirement of MIL-PRF-19500, appendix A.

3.5 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, figure 2, and figure 3.

3.5.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I.

3.7 Electrical test requirements. The electrical test requirements shall be group A as specified herein.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500.

NOTE: The EG1 version in this specification is a construction identifier. It denotes a type of design and not a change in package outline. EG1 is not intended to be a new part number or a new package outline identifier. The EG1 version meets all of the requirements of a non-EG1 version including case outline, hermeticity, and electrical parameters. The EG1 may be marked and shipped as a non-EG1 part number because it is an equivalent device in terms of form, fit, and function. The EG1 is interchangeable and substitutable for the non-EG1 version.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot to this revision to maintain qualification.

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4.2.1.1 JANHC die. Qualification shall be in accordance with appendix H of MIL-PRF-19500.

* 4.3 Screening (JANTX, and JANTXV levels only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table E-IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
(1) 3c	See 4.3.2.
7a 7b	Not applicable Optional
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.1
(2) 13	Subgroup 2 of table I herein. $\Delta V_{F1} \leq +.1, -.2$ V dc; $\Delta I_{R1} \leq 100$ percent of initial values or ±100 nA dc, whichever is greater. Scope display evaluation (see 4.5.3).
14a (3) 14b	Not applicable Required

(1) Shall be performed anytime after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.
(2) Thermal impedance is not required in screen 13, if it has already previously performed.
(2) For clear glass diodes, the hermetic seal (gross leak) may be performed at anytime after temperature cycling.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.2): $I_{O(min)}$ = Rated I_O (see 1.3 herein). T_A = 55°C maximum. Test conditions in accordance with method 1038 of MIL-STD-750, condition B. Adjust I_O or T_A to achieve the required T_J. T_J = 145°C minimum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 as applicable of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (V_C and V_H where appropriate). Measurement delay time (t_{MD}) = 70 μs max. See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I.

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* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIb (JANTX, and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

* 4.4.2.1 Group B inspection, appendix E, table E-VIb, of MIL-PRF-19500.

Subgroup	Method	Inspection
B2	4066	I_{FSM} = 25 A (pk), 10 surges of 1/120 s each at 1 minute intervals, super-imposed on I_0 = 1.0 A dc, V_R = rated V_{RWM} (see 1.3).
B3	1027	I_0 = 1.0 A dc minimum; T_A = +55°C (max.); T_J = +150°C minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3). Adjust I_O to achieve a T_J = +150°C minimum.
B5	4081	+25°C ≤ T_R ≤35°C , $R_{\Theta JL}$ ≤ 42°C/W, L = 3/8 inch.(see 1.3 and 4.3.2).

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with of table I, group A, subgroup 2 herein.

* 4.4.3.1 Group C inspection, appendix E, table E-VII, of MIL-PRF-19500.

Subgroup	Method	Inspection
C2	2036	Terminal strength (lead fatigue), test condition E.
C6	1026	I_0 = 1.0 A dc minimum; T_A = +55°C (max.); T_J = +150°C minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3). Adjust I_O to achieve a T_J = +150°C minimum.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and table II herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750.

* 4.5.2 Free air power burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period. Use method 3100 of MIL-STD-750 to measure T_J.

4.5.2.1 Mounting configurations. Any clips or heat sink mounting configurations may be utilized provided that I_O is adjusted such that the junction temperature of each diode is maintained at T_J = +135°C minimum for burn-in screening and T_J = +150°C minimum for JANTX and JANXV life testing.

4.5.3 Scope display evaluation. Scope display evaluation shall be sharp and stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, group A, subgroup 4 shall be performed on a scope. Reverse current over the knee shall be 500 µA peak.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2 2/						
Thermal impedance	3101	See 4.3.2.	$Z_{\theta JX}$			°C/W
Forward voltage	4011	I_F = 3.0 A (pk), duty cycle ≤ 2 percent pulsed, tp ≤ 8.3 ms	V_{F1}	0.6	1.3	V (pk)
Reverse current leakage	4016	DC method; V_R = rated (see 1.3)	I_{R1}		1.0	µA dc
Breakdown voltage	4021	I_R = 100 µA dc	$V_{(BR)}$			
1N4245, EG1					240	V
1N4246, EG1					480	V
1N4247, EG1					720	V
1N4248, EG1					960	V
1N4249, EG1					1,150	V
Subgroup 3						
High temperature operation:		T_A = +150°C				
Reverse current leakage	4016	DC method, V_R = rated (see 1.3)	I_{R2}		150	µA dc
Low temperature operation:		T_A = -65°C				
Forward voltage	4011	I_F = 3.0 A (pk), duty cycle ≤ 2 percent pulsed, tp ≤ 8.3 ms	V_{F2}		1.5	V dc

See footnotes at end of table.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Reverse recovery time	4031	Condition B, I_F = 0.5 A, I_R = 1.0 A, $I_{(REC)}$ = 0.25 A	t_{rr}		5	µs
Scope display	4023	Sharp and stable, n = 116; c = 0				

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurement only:
Group B, subgroup 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2.
Group E, subgroup 1.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock	1056	100 cycles, 0°C to 100°C.	
Hermetic seal	1071	Test condition C and H1	
Electrical measurement		See table I, group A, subgroup 2.	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A, 1,000 hours.	
Electrical measurement		See table I, subgroup 2.	
Subgroup 4			Sample size = N/A
Thermal impedance, curves		See MIL-PRF-19500.	
Subgroup 5			22 devices c = 0
Barometric pressure (reduced)	1001	c = 0 Pressure = 8 mm Hg for 1N4245, EG1, 1N4246, EG1, 1N4247, EG1. Pressure = 33 mmHg for 1N4248, EG1, 1N4249, EG1.	

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.5.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN). This information in no way implies that manufacturer's PINs are suitable as a substitute for the military PIN.

6.5 Suppliers of die (if applicable).

JANC ordering information	
	Manufacturer
PIN	14552
1N4245, EG1	JANHCA4245
1N4246, EG1	JANHCA4246
1N4247, EG1	JANHCA4247
1N4248, EG1	JANHCA4248
1N4249, EG1	JANHCA4249

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* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19, 99

Preparing activity:
DLA - CC

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